2020 – A1

1. Consider $H = \{h_{\theta_1} : \mathbb{R} \to \{0,1\}, h_{\theta_1}(x) = 1_{[x \geq \theta_1]}(x) = 1_{[\theta_1,+\infty)}(x), \theta_1 \in \mathbb{R}\} \cup \{h_{\theta_2}(x) = 1_{[x<\theta_2]}(x) = 1_{(-\infty,\theta_2)}(x), \theta_2 \in \mathbb{R}\}$. Compute $VCdim(H)$.

2. Consider $\mathbb{H}$ to be the class of all centered in origin sphere classifiers in the 3D space. A centered in origin sphere classifier in the 3D space is a classifier h_r that assigns the value 1 to a point if and only if it is inside the sphere with radius r¿1 and center given by the origin $\mathbf{O}(0,0,0)$. Consider the realizability assumption.

   (a) Show that the class $\mathbb{H}$ can be (ϵ, δ) - PAC learned by giving the algorithm A and determining the sample complexity $m_H(\epsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

   (b) Compute $VCdim(H)$

3. Let $H = \{h_\theta : \mathbb{R} \to \{0,1\}, h_\theta(x) = 1_{[\theta,\theta+1] \cup [\theta+2,+\infty)}(x), \theta \in \mathbb{R}\}$. Compute $VCdim(H)$.

4. An axis aligned square classifier in the plane is a classifier that assigns the value 1 to a point if and only if it is inside a certain square. Formally, given the real numbers $a_1, a_2, r > 0 \in \mathbb{R}$ we define the classifier $h_{(a_1,a_2,r)}$ by

$$h_{(a_1,a_2,r)}(x_1,x_2) = \begin{cases} 1, if a_1 \leq x_1 \leq a_1 + r, a_2 \leq x_2 \leq a_2 + r \\ 0, otherwise \end{cases}$$

   The class of all axis aligned squares in the plane is defined as

$$H = \{h_{(a_1,a_2,r)} : \mathbb{R}^2 \to \{0,1\} | a_1, a_2, r \in \mathbb{R}, r > 0\}$$

   Consider the realizability assumption.

   (b) Find the sample complexity $m_H(\epsilon, \delta)$ in order to show that H is PAC-learnable.

   (c) Compute $VCdim(H)$.

5. Compute the VC-dimension of the class of convex d-gons(convex polygons with exactly d sides) in the plane. Provide a detailed proof of your result.

2020 – A2

1. Consider $H = \{h_{\theta_1} : \mathbb{R} \to \{0,1\}, h_{\theta_1}(x) = 1_{[x \geq \theta_1]}(x) = 1_{[\theta_1,+\infty)}(x) \theta_1 \in \mathbb{R}\} \cup \{h_{\theta_2}(x) = 1_{[x<\theta_2]}(x) = 1_{(-\infty,\theta_2)}(x), \theta_2 \in \mathbb{R}\}$

   a. Compute the shattering coefficient $\tau_H(m)$ of the growth function for $m \geq 0$.

   b. Compare your result with the general upper bound for the growth functions.

2. Let Σ be a finite alphabet and let $\mathbb{X} = \Sigma^m$ be a sample space of all strings of length m over σ. Let $\mathbb{H}$ be a hypothesis space over $\mathbb{X}$, where $\mathbb{H} = \{h_w : \Sigma^m \to \{0,1\}, w \in \Sigma^*, 0 < |w| \leq m, s.t. h_w(x) = 1$ if w is a substring of $x\}$.

   a. Give an upper bound(any upper bound that you can come up with) of the VC-dimension of $\mathbb{H}$ in terms of $|\Sigma|$ and m. In order to find an upper bound

   b. Give an efficient algorithm for finding a hypothesis h_w consistent with a training set in the realizable case. What is the complexity of your algorithm?

3. Consider the boosting algorithm described(page 4) in the article "Rapid object detection using a boosted cascade of simple features", P. Viola and M. Jones, CVPR 2001. Consider that the number of positives is equal with the number of negative examples(l=m).

   a. Prove that the distribution w_{t+1} obtained at round $t+1$ based on the algorithm described in the article is the same with the distribution $\mathbf{D}^{(t+1)}$ obtained on the procedure described in lecture 11(slides 10-12).

   b. Prove that the two final classifiers(the one described in the article and the one described in the lecture) are equivalent.

2021 – A1

Exercise 1

Statement:

Give an example of a finite hypothesis class $\mathcal{H}$ with $VCdim(\mathcal{H}) = 2021$. Justify your choice.

Exercise 2

Statement:

Consider $\mathcal{H}_{balls}$ to be the set of all balls in $\mathbf{R}^2$:

$\mathcal{H}_{balls} = \left\{ B(x, r), x \in \mathbb{R}^2, r \geq 0 \right\}$, where $B(x, r) = \left\{ y \in \mathbb{R}^2 \mid \|y - x\|_2 \leq r \right\}$

As mentioned in the lecture, we can also view $\mathcal{H}_{balls}$ as the set of indicator functions of the balls $B(x, r)$ in the plane: $\mathcal{H}_{balls} = \left\{ h_{x,r} : \mathbb{R}^2 \longrightarrow \{0, 1\}, h_{x,r} = \mathbf{1}_{B(x,r)}, x \in \mathbb{R}^2, r > 0 \right\}$.

Can you give an example of a set A in $\mathbf{R}^2$ of size 4 that is shattered by $\mathcal{H}_{balls}$? Give such an example or justify why you cannot find a set A of size 4 shattered by $\mathcal{H}_{balls}$.

Exercise 3

Statement:

Let $\mathcal{X} = \mathbf{R}^2$ and consider $\mathcal{H}_\alpha$ the set of concepts defined by the area inside a right triangle ABC with two catheti AB and AC parallel to the axes (O_x and O_y) and with $AB/AC = \alpha$ (fixed constant > 0). Consider the realizability assumption. Show that the class $\mathcal{H}_\alpha$ can be (ε, δ) - PAC learned by giving an algorithm A and determining an upper bound on the sample complexity $m_{\mathcal{H}}(\varepsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

Exercise 4

Statement:

Consider $\mathcal{H}$ to be the class of all centered in origin sphere classifiers in the 3D space. A centered in origin sphere classifier in the 3D space is a classifier h_r that assigns value 1 to a point if and only if it is inside the sphere with radius $r > 0$ and center given by the origin $\mathbf{O}(0, 0, 0)$. Consider the realizability assumption.

a. show that the class $\mathcal{H}$ can be (ε, δ) - PAC learned by giving an algorithm A and determining an upper bound on the sample complexity $m_{\mathcal{H}}(\varepsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

b. compute VCdim($\mathcal{H}$).

Exercise 5

Statement:

Consider $\mathcal{H} = \left\{ h_\theta : \mathbb{R} \longrightarrow \{0, 1\} \mid h_\theta(x) = \mathbf{1}_{[\theta, \theta+1] \cup [\theta+2, \infty)}, \theta \in \mathbb{R} \right\}$. Compute $VCdim(\mathcal{H})$.

2021 – A2

Exercise 1

Statement:

Consider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2$ where

$\mathcal{H}_1 = \{h_{\theta_1} : \mathbb{R} \longrightarrow \{0, 1\},\ h_{\theta_1}(x) = \mathbf{1}_{[x \geq \theta_1]}(x) = \mathbf{1}_{[\theta_1, +\infty)}(x),\ \theta_1 \in \mathbb{R}\}$ and

$\mathcal{H}_2 = \{h_{\theta_2} : \mathbb{R} \longrightarrow \{0, 1\},\ h_{\theta_2}(x) = \mathbf{1}_{[x < \theta_2]}(x) = \mathbf{1}_{(-\infty, \theta_2)}(x),\ \theta_2 \in \mathbb{R}\}$

a) Give an efficient ERM algorithm for learning $\mathcal{H}$ and compute its complexity for the realizable case.

b) Give an efficient ERM algorithm for learning $\mathcal{H}$ and compute its complexity for the agnostic case.

c) Compute the shattering coefficient $\tau_H(m)$ of the growth function for m ≥ 0 for hypothesis class $\mathcal{H}$.

d) Compare your result with the general upper bound for the growth functions and show that $\tau_H(m)$ obtained at previous point c is not equal to the upper bound.

e) Does there exist a hypothesis class $\mathcal{H}$ for which $\tau_H(m)$ is equal to the general upper bound (over $\mathbb{R}$ or another domain χ)? If your answer is yes please provide an example, if your answer is no please provide a justification.

2021 – A2

Exercise 2

Statement:

Consider a modified version of the AdaBoost algorithm that runs for exactly three rounds as follows:

- the first two rounds run exactly as in AdaBoost (at round 1 we obtain distribution $\mathbf{D}^{(1)}$, weak classifier h_1 with error ε_1, at round 2 we obtain distribution $\mathbf{D}^{(2)}$, weak classifier h_2 with error ε_2).
- in the third round we compute for each i = 1, 2, …, m:

$$D^3(i) = \begin{cases} \frac{D^{(1)}(i)}{Z}, & if\ h_1(x_i) \neq h_2(x_i), \\ 0, & otherwise \end{cases}$$

  where Z is a normalization factor such that $\mathbf{D}^{(3)}$ is a probability distribution.
- obtain weak classifier h_3 with error ε_3.
- output the final classifier $h_{final}(x) = sign(h_1(x) + h_2(x) + h_3(x))$.

Assume that at each round *t = 1, 2, 3* the weak learner returns a weak classifier h_t for which the error ε_t satisfies $\varepsilon_t \leq 1/2 - \gamma_t$, $\gamma_t > 0$.

a) What is the probability that the classifier h_1 (selected at round 1) will be selected again at round 2? Justify your answer.

b) Consider $\gamma = \min\{\gamma_1, \gamma_2, \gamma_3\}$. Show that the training error of the final classifier h_{final} is at most $\frac{1}{2} - \frac{3}{2}\gamma + 2\gamma^3$ and show that this is strictly smaller than $\frac{1}{2} - y$.

2021 – A2

Exercise 3

Statement:

Let Σ be a finite alphabet and let $\chi = \Sigma^m$ be a sample space of all strings of length m over Σ. Let $\mathcal{H}$ be a hypothesis space over χ, where $\mathcal{H} = \{h_w : \Sigma^m \longrightarrow \{0,1\},\ w \in \Sigma^*,\ 0 < |w| \leq m,$ s.t $h_w(x) = 1$ if w is a substring of x$\}$.

a. Give an upper bound (any upper bound that you can come up) of the VCdimension of $\mathcal{H}$ in terms of $|\Sigma|$ and m.

b. Give an efficient algorithm for finding a hypothesis h_w consistent with a training set in the realizable case. What is the complexity of your algorithm?

Example: let Σ = {a, b, c}, m = 4 and the training set S = {(aabc, 1), (baca, 0), (bcac,0), (abba, 1)}. The output of the algorithm should be h_{ab}.

2022 – A1

1 Exercise 1

1. **(0.5 points)** Give an example of a finite hypothesis class $\mathcal{H}$ with VCdim($\mathcal{H}$) = 2022. Justify your choice.

3 Exercise 2

2. **(0.5 points)** What is the maximum value of the natural even number n, $n = 2m$, such that there exists a hypothesis class $\mathcal{H}$ with n elements that shatters a set C of $m = \frac{n}{2}$ points? Give an example of such an $\mathcal{H}$ and C. Justify your answer.

5 Exercise 3

3. **(0.75 points)** Let $\mathcal{X} = R^2$ and consider $\mathcal{H}$ the set of axis aligned rectangles with the center in origin O(0, 0). Compute the $VCdim(\mathcal{H})$.

7 Exercise 4

4. **(1 point)** Let $\mathcal{X} = R^2$ and consider $\mathcal{H}_\alpha$ the set of concepts defined by the area inside a right triangle ABC with two catheti AB and AC parallel to the axes (Ox and Oy), and with the ratio AB/AC = α (fixed constant > 0). Consider the realizability assumption. Show that the class $\mathcal{H}_\alpha$ is (ϵ, δ)-PAC learnable by giving an algorithm A and determining an upper bound on the sample complexity $m_H(\epsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

9 Exercise 5

5. **(1.25 points)** Consider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$, where:

$$\mathcal{H}_1 = \{h_{\theta_1} : R \to \{0,1\} \mid h_{\theta_1}(x) =_{[x \geq \theta_1]} (x) =_{[\theta_1, +\infty)} (x), \theta_1 \in R\},$$

$$\mathcal{H}_2 = \{h_{\theta_2} : R \to \{0,1\} \mid h_{\theta_2}(x) =_{[x < \theta_2]} (x) =_{(-\infty, \theta_2)} (x), \theta_2 \in R\},$$

$$\mathcal{H}_3 = \{h_{\theta_1,\theta_2} : R \to \{0,1\} \mid h_{\theta_1,\theta_2}(x) =_{[\theta_1 \leq x \leq \theta_2]} (x) =_{[\theta_1, \theta_2]} (x), \theta_1, \theta_2 \in R\}.$$

2022 – A1

11 Exercise 6

6. **(1 point)** A decision list may be thought of as an ordered sequence of if-then-else statements. The sequence of conditions in the decision list is tested in order, and the answer associated with the first satisfied condition is output.

More formally, a *k-decision list* over the boolean variables $x_1, x_2, \ldots, x_n$ is an ordered sequence $L = \{(c_1, b_1), (c_2, b_2), \ldots, (c_l, b_l)\}$ and a bit b, in which each c_i is a conjunction of at most k literals over $x_1, x_2, \ldots, x_n$ and each $b_i \in \{0, 1\}$. For any input $a \in \{0,1\}^n$, the value $L(a)$ is defined to be b_j where j is the smallest index satisfying $c_j(a) = 1$; if no such index exists, then $L(a) = b$. Thus, b is the "default" value in case a falls off the end of the list. We call b_i the bit associated with the condition c_i.

The next figure shows an example of a *2-decision list* along with its evaluation on a particular input.

Show that the VC dimension of 1-decision lists over $\{0,1\}^n$ is lower and upper bounded by linear functions, by showing that there exists $\alpha, \beta, \gamma, \delta \in R$ such that:

$$\alpha \cdot n + \beta \leq VCdim(\mathcal{H}_{1-decision\ list}) \leq \gamma \cdot n + \delta$$

Hint: Show that 1-decision lists over $\{0,1\}^n$ compute linearly separable functions (halfspaces).

2022 – A2

1. **(1.5 points)** Consider $\mathcal{H}$ the class of 3-piece classifiers (signed intervals):

$$\mathcal{H} = \{h_{a,b,s} : \mathbb{R} \to \{-1,1\} \mid a \le b, s \in \{-1,1\}\}, \text{ where } h_{a,b,s}(x) = \begin{cases} s, & x \in [a,b] \\ -s, & x \notin [a,b] \end{cases}$$

   a. Compute the shattering coefficient $\tau_H(m)$ of the growth function for $m \ge 0$ for hypothesis class $\mathcal{H}$. **(1 point)**

   b. Compare your result with the general upper bound for the growth functions and show that $\tau_H(m)$ obtained at previous point a is not equal with the upper bound. **(0.25 points)**

   c. Does there exist a hypothesis class $\mathcal{H}$ for which is equal to the general upper bound (over or another domain $\mathcal{X}$)? If your answer is yes please provide an example, if your answer is no please provide a justification. **(0.25 points)**

2. **(1.5 points)** Consider the concept class C_2 formed by the union of two closed intervals $[a,b] \cup [c,d]$, where $a,b,c,d \in \mathbb{R}, a \le b \le c \le d$. Give an efficient ERM algorithm for learning the concept class C_2 and compute its complexity for each of the following cases:

   a. realizable case. **(1 point)**

   b. agnostic case. **(0.5 point)**

3. **(1j.5 points)** Consider a modified version of the AdaBoost algorithm that runs for exactly three rounds as follows:

   - the first two rounds run exactly as in AdaBoost (at round 1 we obtain distribution $\mathbf{D}^{(1)}$, weak classifier h_1 with error ϵ_1; at round 2 we obtain distribution $\mathbf{D}^{(2)}$, weak classifier h_2 with error ϵ_2).
   - in the third round we compute for each $i = 1, 2, \ldots, m$:

$$\mathbf{D}^{(3)}(i) = \begin{cases} \frac{D^{(1)}(i)}{Z}, & \text{if } h_1(x_i) \ne h_2(x_i) \\ 0, & \text{otherwise} \end{cases}$$

     where Z is a normalization factor such that $\mathbf{D}^{(3)}$ is a probability distribution.

   - obtain weak classifier h_3 with error ϵ_3.

2022 – A2

   - output the final classifier $h_{final}(x) = sign(h_1(x) + h_2(x) + h_3(x))$.

   Assume that at each round $t = 1, 2, 3$ the weak learner returns a weak classifier h_t for which the error ϵ_t satisfies $\epsilon_t \le \frac{1}{2} - \gamma_t, \gamma_t > 0$.

   a. What is the probability that the classifier h_1 (selected at round 1) will be selected again at round 2? Justify your answer. **(0.75 points)**

   b. Consider $\gamma = min\{\gamma_1, \gamma_2, \gamma_3\}$. Show that the training error of the final classifier h_{final} is at most $\frac{1}{2} - \frac{3}{2}\gamma + \gamma^2$ and show that this is strictly smaller than $\frac{1}{2} - \gamma$. **(0.75 points)**

4. **(1 point)** Consider H^d_{2DNF} the class of 2-term disjunctive normal form formulae consisting of hypothesis of the form $h : \{0,1\}^d \to \{0,1\}$,

$$h(x) = A_1(x) \vee A_2(x)$$

where $A_i(x)$ is a Boolean conjunction of literals H^d_{conj}.

It is known that the class H^d_{2DNF} is not efficient properly learnable but can be learned improperly considering the class H^d_{2CNF}. Give a γ-weak-learner algorithm for learning the class H^d_{2DNF} which is not a stronger PAC learning algorithm for H^d_{2DNF} (like the one considering H^d_{2CNF}). Prove that this algorithm is a γ-weak-learner algorithm for H^d_{2DNF}.

Hint: Find an algorithm that returns $h(x) = 0$ or the disjunction of 2 literals.

2023 – A1

1. **(0.75 points)** Give examples for:

a) a finite hypothesis class $\mathcal{H}$ with VCdim($\mathcal{H}$) = 2023. Justify your choice. **(0.25 points)**

b) an infinite hypothesis class $\mathcal{H}$ with VCdim($\mathcal{H}$) = 2023. Justify your choice. **(0.25 points)**

c) an infinite hypothesis class $\mathcal{H}$ with VCdim($\mathcal{H}$) = ∞. Justify your choice. **(0.25 points)**

2. **(0.75 points)** Consider $\mathcal{H}$ to be the following hypothesis class:

$$\mathcal{H} = \{h_a : \mathbb{R}^3 \to \{0,1\} \mid h_a(\mathbf{x}) = \mathbb{1}_{[\|x\|_2 \leq a]}(\mathbf{x}), \mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3, \|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2}\}$$

Consider the realizability assumption.

a) show that $\mathcal{H}$ can be (ϵ,δ)-PAC learned by giving an algorithm A and determining the sample complexity $m_{\mathcal{H}}(\epsilon,\delta)$ such that the definition of PAC learnability is satisfied. **(0.5 points)**

b) compute VCdim($\mathcal{H}$). **(0.25 points)**

3. **(1 point)** Compute VCdim($\mathcal{H}$) where $\mathcal{H}$ is the following hypothesis class:

$$\mathcal{H} = \{h_{\theta_1,\theta_2} : \mathbb{R}^2 \to \{0,1\} \mid h_{\theta_1,\theta_2}(\mathbf{x}) = h_{\theta_1,\theta_2}(x_1,x_2) = \mathbb{1}_{[\theta_1 + x_1 \cdot sin(\theta_2) + x_2 \cdot cos(\theta_2) > 0]}, \theta_1, \theta_2 \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^2\}.$$

4. **(1.5 points)** Consider $\mathcal{H}_\alpha$ to be the class of axis aligned rectangles with fixed *aspect-ratio* α, where $\alpha \in \mathbb{R}$ and $\alpha > 0$:

$$\mathcal{H}_\alpha = \{h_{a,b,c,d} : \mathbb{R}^2 \to \{0,1\} \mid h_{a,b,c,d}(\mathbf{x}) = \mathbb{1}_{[a,b]\times[c,d]}(\mathbf{x}), a,b,c,d \in \mathbb{R}, a < b, c < d, \frac{d-c}{b-a} = \alpha\}$$

Consider the realizability assumption.

a) give a learning algorithm A that returns a hypothesis h_S from $\mathcal{H}_\alpha$, $h_S = A(S)$ consistent with the training set S (h_S has empirical risk 0 on S). **(0.75 points)**

b) find the sample complexity $m_{\mathcal{H}_\alpha}(\epsilon,\delta)$ in order to show that $\mathcal{H}_\alpha$ is PAC – learnable. **(0.75 points)**

5. **(1 point)** Compute VCdim($\mathcal{H}$) where $\mathcal{H}$ is the following hypothesis class:

$$\mathcal{H} = \{h_\theta : \mathbb{R} \to \{0,1\} \mid h_\theta(x) = \mathbb{1}_{[\theta,\theta+1]\cup[\theta+2,\theta+4]\cup[\theta+6,\theta+9]}(x), \theta \in \mathbb{R}\}.$$

2023 – A2

1. **(1.25 points)** Consider $\mathcal{H}$ the following hypothesis class :

$$\mathcal{H} = \{h_a : \mathbb{R} \to \{0,1\} \mid a > 0, a \in \mathbb{R}, \text{ where } h_a(x) = \mathbb{1}_{[-a,a]}(x) = \begin{cases} 1, & x \in [-a,a] \\ 0, & x \notin [-a,a] \end{cases} \}$$

a. Compute the growth function $\tau_H(m)$ (also known as shatter coefficient) for $m \geq 0$ for hypothesis class $\mathcal{H}$. **(1 point)**

b. Compare your result from the previous point with the general upper bound given by the Sauer lemma. Are they equal or different? **(0.25 points)**

2. **(1.5 points)** Consider de concept class C_a formed by the union of two closed intervals $[a, a+1] \cup [a+2, a+4]$, where $a \in \mathbb{R}$. Give an efficient ERM algorithm for learning the concept class C_a and compute its complexity for each of the following cases:

a. realizable case. **(1 point)**

b. agnostic case. **(0.5 point)**

Solution:

3. **(1.25 points)** Consider a modified version of the AdaBoost algorithm that runs for exactly three rounds as follows:

- the first two rounds run exactly as in AdaBoost (at round 1 we obtain distribution $\mathbf{D}^{(1)}$, weak classifier h_1 with error ϵ_1; at round 2 we obtain distribution $\mathbf{D}^{(2)}$, weak classifier h_2 with error ϵ_2).
- in the third round we compute for each $i = 1, 2, \ldots, m$:

$$\mathbf{D}^{(3)}(i) = \begin{cases} \frac{D^{(1)}(i)}{Z}, & \text{if } h_1(x_i) \neq h_2(x_i) \\ 0, & \text{otherwise} \end{cases}$$

where Z is a normalization factor such that $\mathbf{D}^{(3)}$ is a probability distribution.

- obtain weak classifier h_3 with error ϵ_3.
- output the final classifier $h_{final}(x) = sign(h_1(x) + h_2(x) + h_3(x))$.

Assume that at each round $t = 1, 2, 3$ the weak learner returns a weak classifier h_t for which the error ϵ_t satisfies $\epsilon_t \leq \frac{1}{2} - \gamma_t, \gamma_t > 0$.

a. What is the probability that the classifier h_1 (selected at round 1) will be selected again at round 2? Justify your answer. **(0.25 points)**

b. Consider $\gamma = min\{\gamma_1, \gamma_2, \gamma_3\}$. Show that the training error of the final classifier h_{final} is at most $\frac{1}{2} - \frac{3}{2}\gamma + 2\gamma^3$ and show that this is strictly smaller than $\frac{1}{2} - \gamma$. **(1 point)**

2023 – A2

4. **(1 point)** Consider H^d_{2DNF} the class of 2-term disjunctive normal form formulae consisting of hypothesis of the form $h : \{0,1\}^d \to \{0,1\}$,

$$h(x) = A_1(x) \vee A_2(x)$$

where $A_i(x)$ is a Boolean conjunction of literals H^d_{conj}.

It is known that the class H^d_{2DNF} is not efficient properly learnable but can be learned improperly considering the class H^d_{2CNF}. Give a γ-weak-learner algorithm for learning the class H^d_{2DNF} which is not a strong PAC learning algorithm for H^d_{2DNF} (like the one considering H^d_{2CNF}). Prove that this algorithm is a γ-weak-learner algorithm for H^d_{2DNF}.

Hint: Find an algorithm that returns $h(x) = 0$ or the disjunction of 2 literals.

2024 – A1

Exercise 1. Give examples for:

a) a finite hypothesis class $\mathcal{H}$ with $\mathrm{VCdim}(\mathcal{H}) = 2024$. Justify your choice.

b) an infinite hypothesis class $\mathcal{H}$ with $\mathrm{VCdim}(\mathcal{H}) = 2024$. Justify your choice.

Exercise 3. **C**onsider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$, where:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \to \{0,1\} \mid h_a(x) = \mathbf{1}_{[x \geq a]}(x) = \mathbf{1}_{[a,+\infty)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_b : \mathbb{R} \to \{0,1\} \mid h_b(x) = \mathbf{1}_{[x<b]}(x) = \mathbf{1}_{(-\infty,b)}(x), b \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_{c,d} : \mathbb{R} \to \{0,1\} \mid h_{c,d}(x) = \mathbf{1}_{[c \leq x \leq d]}(x) = \mathbf{1}_{[c,d]}(x), c, d \in \mathbb{R}\}.$$

Consider the realizability assumption. Compute $\mathrm{VCdim}(\mathcal{H})$.

Exercise 4. Consider $\mathcal{H}$ the class of 3-piece classifiers (signed intervals):

$$\mathcal{H} = \{h_{a,b,s} : \mathbb{R} \to \{-1,1\} \mid a \leq b, s \in \{-1,1\}\}, \text{ where } h_{a,b,s}(x) = \begin{cases} s, & x \in [a,b] \\ -s, & x \notin [a,b] \end{cases}$$

a. Compute the shattering coefficient $\tau_H(m)$ of the growth function for $m \geq 0$ for hypothesis class $\mathcal{H}$.

b. Compare your result with the general upper bound for the growth functions and show that $\tau_H(m)$ obtained at previous point a is not equal with the upper bound.

c. Does there exist a hypothesis class $\mathcal{H}$ for which the shattering coefficient $\tau_H(m)$ is equal to the general upper bound (over or another domain $\mathcal{X}$)? If your answer is yes please provide an example, if your answer is no please provide a justification.

Excercise 5. Let $\mathcal{H} = \{h : \mathbb{R} \to \{0,1\} \mid h_\theta(x) = \mathbf{1}_{[\theta,\theta+1] \cup [\theta+2,\infty)}(x), \theta \in \mathbb{R}\}$.
Compute $\mathrm{VCdim}(\mathcal{H})$.

Advanced Machine Learning
Exam, 21^{st} of June 2024

In the following, we denote with $\mathbf{1}_A(x)$ the indicator function of the set A. $\mathbf{1}_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$

Exercise 1. (1 point) Give examples of two hypothesis classes $\mathcal{H}_1$ and $\mathcal{H}_2$ such that:

a. $VCdim(\mathcal{H}_1 \cup \mathcal{H}_2) = max(VCdim(\mathcal{H}_1), VCdim(\mathcal{H}_2))$ **(0.5 points)**

b. $VCdim(\mathcal{H}_1 \cup \mathcal{H}_2) > max(VCdim(\mathcal{H}_1), VCdim(\mathcal{H}_2))$ **(0.5 points)**

Justify your answers.

Exercise 2. (1.5 points) Let $\mathcal{H}_n$, with $n \geq 1$, be the following hypothesis class:

$$\mathcal{H}_n = \{h_a : \mathbb{R}^n \to \{0,1\} | h_a(\mathbf{x}) = \mathbf{1}_{[0,a]}(\|\mathbf{x}\|_2), a \in \mathbb{R}, a > 0\}, \textit{where}$$

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \ldots x_n^2}, \mathbf{x} = (x_1, x_2, \ldots, x_n)$$

a. Compute the VC-dimension of $\mathcal{H}_n$ for $n \geq 1$. **(1 point)**

b. Compute the shattering coefficient $\tau_{\mathcal{H}_n}(m)$ of the growth function for $m \geq 0$. **(0.5 points)**

Exercise 3. (1 point) Consider de concept class C given by the closed intervals $[a, a+1]$, where $a \in \mathbb{R}$. Give an efficient ERM algorithm for learning the concept class C and compute its complexity for each of the following cases:

a. realizable case. **(0.5 points)**

b. agnostic case. **(0.5 points)**

Exercise 4. (1.5 points) Let $\mathcal{X}$ be an instance space and consider $\mathcal{H} \subseteq \{0,1\}^{\mathcal{X}}$ to be a hypothesis space with finite VC-dimension. For each $x \in \mathcal{X}$, we consider the function $z_x : \mathcal{H} \to \{0,1\}$ such that we have $z_x(h) = h(x)$ for each $h \in \mathcal{H}$. Let $Z = \{z_x : H \to \{0,1\}, x \in \mathcal{X}\}$. Prove that $VCdim(Z) < 2^{VCdim(H)+1}$.

Ex-officio: 0.5 points.

Time: 3 hours.